

GCP Data Engineering Tool Map

Think of GCP as a toolbox for this lifecycle:

Source systems

→ Ingestion → Storage → Processing / transformation → Analytics / ML → Orchestration → Governance / security → Monitoring / reliability → AI agents / automation

Data lake = cheap/flexible raw file storage

Data warehouse = clean, structured, fast SQL analytics

Lakehouse = data lake flexibility + warehouse-style analytics/governance

1. Storage and warehouse tools

Cloud Storage

Use this when you need cheap, scalable object storage.

Examples:

Raw CSVs

JSON logs

Images

Backups

Data lake landing zone

Files from vendors

It is usually the **raw storage layer**.

BigQuery

Use this when you need a scalable analytics warehouse.

Examples:

SQL analytics

Dashboard tables

Reporting datasets

Cleaned business tables

Ad hoc analyst queries

Large-scale aggregations

BigQuery is where curated, queryable analytics data usually lives.

BigLake

Use this when data is physically in Cloud Storage or another object store, but you want BigQuery-style governance and access control.

Examples:

- Query Parquet files in Cloud Storage
- Govern lakehouse data
- Apply fine-grained access controls
- Avoid copying all lake data into BigQuery

BigLake is for governed (rule and controls) lakehouse access. It lets me keep data in object storage but manage/query it more like warehouse data.

External tables

Use this when you want BigQuery to query files directly from Cloud Storage without loading them into native BigQuery storage.

Examples:

- Quickly query CSV/Parquet files in GCS
- Avoid loading temporary data
- Explore raw data

External tables are useful for exploration or when I do not want to duplicate data into BigQuery.

Difference:

- External table = query external files
- BigLake table = external table with stronger governance/security/lakehouse features
- Native BigQuery table = data stored inside BigQuery

2. Ingestion tools

bq load / LOAD DATA / BigQuery UI load

Use these when you need to load files into BigQuery.

Examples:

- Load CSV from Cloud Storage into BigQuery
- Load local CSV through UI
- Load Parquet/Avro/JSON into warehouse

Job logic:

“Use load jobs for simple batch ingestion into BigQuery.”

BigQuery Data Transfer Service

Use this when you want scheduled, managed transfers into BigQuery.

Examples:

Google Ads → BigQuery
YouTube reports → BigQuery
Scheduled SaaS exports
Recurring data movement

BigQuery Data Transfer Service is the managed scheduling and automating for transferring data into BigQuery from supported sources.

Pub/Sub

Use this for event ingestion and messaging.

Examples:

App click events
Order events
IoT sensor events
Payment events
Streaming logs

Job logic:

“Pub/Sub is the front door for streaming/event-driven data.”

Pub/Sub does not usually transform your data. It receives and distributes messages.

Datastream

Use this when you need change data capture, also called CDC.

Examples:

Replicate MySQL changes to BigQuery
Replicate PostgreSQL changes
Near-real-time database sync

If I need changes from an operational database copied continuously into analytics systems, I think Datastream. It can transfer and capture both metadata and regular data

Storage Transfer Service

Use this when you need to move files into Cloud Storage.

Examples:

AWS S3 → Cloud Storage

On-prem files → Cloud Storage

Scheduled bucket-to-bucket transfer

Job logic:

“Storage Transfer moves files. BigQuery Data Transfer moves data into BigQuery.”

3. Processing and transformation tools

Dataflow

Use this for serious batch (Scheduled time everyday for example) or streaming processing (continuous input of data). Dataflow is Google’s managed Apache Beam service for batch and streaming pipelines, real-time analytics, ETL, and complex transformations.

Examples:

Pub/Sub → Dataflow → BigQuery

Clean streaming events

Window events by time

Deduplicate messages

Enrich records

Batch transform large files

Job logic:

If I need scalable transformations, especially streaming, Dataflow is the default.

Dataproc

Use this for Spark/Hadoop workloads.

Examples:

Company already has Spark jobs

PySpark ETL
Hadoop migration
Large distributed processing

Job logic:

“If the team already uses Spark, or the code is Spark-based, Dataproc is the GCP-managed option.”

Google’s analytics decision guidance describes Dataproc as the managed Hadoop/Spark option and Data Fusion as a point-and-click ingestion/ETL tool with many connectors.

([Google Cloud](#))

Data Fusion

Use this for low-code/no-code ETL pipelines.

Examples:

Drag-and-drop ETL
Connector-heavy ingestion
Business-friendly pipeline building
Simple transformations without writing Beam/Spark code

Job logic:

“If the team wants visual ETL instead of code-first pipelines, Data Fusion is the option.”

Dataform

Use this for SQL-based transformation workflows in BigQuery.

Examples:

Raw BigQuery tables → cleaned tables
Analytics engineering
dbt-style SQL pipelines
Dependency management between SQL models

Job logic:

“If transformations are mostly SQL inside BigQuery, Dataform is the clean way to manage them.”

This matters a lot for modern analytics engineering.

BigQuery scheduled queries

Use this for simple recurring SQL transformations.

Examples:

Run SQL every morning
Create daily summary table
Refresh reporting table

Job logic:

“If it is just one or two SQL jobs, scheduled queries are simpler than Composer/Dataform.”

4. Orchestration tools

Cloud Composer

Use this when you need workflow orchestration. It is Google’s managed Apache Airflow.

Examples:

Run extract → transform → load → quality checks → notify
Schedule many dependent jobs
Coordinate BigQuery, Dataflow, Dataproc, APIs

Job logic:

“Composer does not do the heavy processing itself. It schedules and coordinates the tools that do.”

Workflows

Use this for lightweight orchestration of APIs and cloud services.

Examples:

Call Cloud Function
Then call BigQuery job
Then send notification

Job logic:

“Use Workflows for simpler API/service orchestration. Use Composer for heavier data DAGs.”

Cloud Scheduler

Use this like cron.

Examples:

Trigger a Cloud Run job every day
Trigger a workflow every hour

Job logic:

“Cloud Scheduler starts something on a schedule. Composer manages complex dependency graphs.”

5. Serving, BI, and ML tools

Looker / Looker Studio

Use these for dashboards and BI.

Examples:

Executive dashboards
Business reporting
Self-service analytics
Metrics layer

Job logic:

“BigQuery stores/query data. Looker visualizes and models business metrics.”

Vertex AI

Use this for machine learning and AI workloads.

Examples:

- Train model
- Deploy model endpoint
- Feature engineering
- Generative AI
- Embeddings
- Agent building
- Model monitoring

Job logic:

“Vertex AI is the main ML/AI platform.”

BigQuery ML

Use this when you want ML directly inside BigQuery using SQL.

Examples:

- Forecasting
- Classification
- Regression
- Clustering
- Anomaly detection

Job logic:

“If the data is already in BigQuery and the model is standard, BigQuery ML is faster than building a full ML pipeline.”

6. Governance, security, and metadata tools

Dataplex / Knowledge Catalog

Use this for governance, metadata, data discovery, lineage, and quality.

Examples:

- Catalog data assets
- Track lineage
- Find who owns a table
- Apply data quality rules
- Organize lake/warehouse assets

Google describes data lineage as a map of the full data lifecycle: where data comes from, where it goes, and what transformations happen along the way. Supported services such as BigQuery and Dataflow can automatically report lineage events when the Data Lineage API is enabled. ([Google Cloud Documentation](#))

Job logic:

“When the problem is trust, discovery, compliance, ownership, or lineage, think Dataplex/Knowledge Catalog.”

IAM

Use this for identity and permissions.

Examples:

- Who can access a project?
- Who can run BigQuery jobs?
- Who can view a bucket?
- Who can deploy a Dataflow job?

Job logic:

“IAM controls who can do what.”

Policy tags / column-level security / row-level security

Use these to restrict sensitive data in BigQuery.

Examples:

- Hide SSNs
- Restrict salary columns
- Let managers see only their region
- Mask PII

Job logic:

“If the data is sensitive, I need access controls at table, row, column, or policy-tag level.”

Cloud DLP / Sensitive Data Protection

Use this to detect, classify, mask, or tokenize sensitive data.

Examples:

Find emails, phone numbers, SSNs
De-identify sensitive fields
Scan buckets/tables for PII

Job logic:

“If the issue is discovering or masking PII, think Sensitive Data Protection.”

7. Monitoring and reliability tools

Cloud Logging

Use this for logs.

Examples:

Pipeline logs
Error messages
Function logs
Dataflow job logs

Cloud Monitoring

Use this for metrics, alerts, dashboards, and uptime.

Examples:

Alert if pipeline fails
Alert if latency spikes
Alert if BigQuery slots/costs are high

Error Reporting / Trace

More relevant for app/backend work, but useful when pipelines have services or APIs.

Job logic:

“A data engineer must not just build pipelines; they must monitor them.”

Phase 2 — Google ADK and Google’s Agent Platform

This is newer and important because data engineering is moving toward AI-assisted workflows.

Google ADK

Google ADK stands for **Agent Development Kit**.

It is Google’s open-source framework for building, debugging, and deploying AI agents. Google describes ADK as an open-source agent development framework for building reliable AI agents at enterprise scale. ([Google Cloud Documentation](#))

The standalone ADK site says it supports Python, TypeScript, Go, Java, and Kotlin. ([Adk](#))

Use ADK when you want to build an agent that can:

- Reason over a task
- Use tools
- Call APIs
- Access data
- Remember context
- Coordinate sub-agents
- Return useful results

Simple example:

User asks: “Why did revenue drop yesterday?”

Agent:

1. Queries BigQuery
2. Checks latest pipeline run

3. Looks at data quality results
4. Summarizes possible causes
5. Creates a Slack/email update

That is not just a chatbot. That is an agent connected to tools.

Gemini Enterprise Agent Platform / Agent Platform

Google's Agent Platform is the broader enterprise platform for building, scaling, governing, and optimizing agents. Google describes Gemini Enterprise Agent Platform as a comprehensive platform for developers to build, scale, govern, and optimize agents. ([Google Cloud](#))

The platform includes pieces like:

- ADK
- Agent Studio
- Agent Garden
- Model Garden
- Agent Runtime / Agent Engine
- Sessions and memory
- Governance and identity
- Evaluation and observability

Google's Agent Platform overview describes ADK as the modular, model-agnostic framework; Agent Studio as a low-code visual canvas; Agent Garden as prebuilt agents/templates; and Model Garden as access to Google, third-party, and open-source models. ([Google Cloud Documentation](#))

The simple difference

- ADK = framework to build agents
- Agent Platform = managed enterprise platform to deploy, scale, govern, and monitor agents
- Agent Studio = visual/low-code agent builder
- Agent Garden = templates/prebuilt agents
- Model Garden = model catalog
- Vertex AI / Gemini models = intelligence layer

For your job, you should understand agents like this:

"An AI agent is useful when a workflow requires reasoning plus tool use, not just answering a question."

In data engineering, agent use cases include:

- Pipeline failure investigation agent
- Data quality monitoring agent
- Business metrics explanation agent
- SQL assistant over BigQuery
- Metadata/catalog search agent
- PII detection and remediation agent

Phase 3 — Important Terminology and Processes

Here are the terms you should know cold.

Ingestion

Moving data from a source into your data platform.

Examples:

- CSV → Cloud Storage
- App events → Pub/Sub
- Postgres → Datastream → BigQuery
- SaaS connector → BigQuery

Batch ingestion

Data arrives on a schedule.

Examples:

- Every hour
- Every night
- Every Monday
- Monthly finance file

Tools:

- Cloud Storage
- BigQuery load

BigQuery Data Transfer Service
Data Fusion
Composer
Storage Transfer Service

Streaming ingestion

Data arrives continuously.

Examples:

Clicks
Orders
IoT events
Fraud events
Log events

Tools:

Pub/Sub
Dataflow
BigQuery streaming
Cloud Run

ETL

Extract → Transform → Load.

Meaning:

Pull data from source
Transform it before warehouse
Load cleaned data

Example:

API → Dataflow cleans data → BigQuery

Use ETL when transformation should happen before storing in the warehouse.

ELT

Extract → Load → Transform.

Meaning:

Pull data

Load raw data first

Transform later inside warehouse

Example:

CSV → Cloud Storage → BigQuery raw table → Dataform SQL clean table

Use ELT when BigQuery is powerful enough to transform after loading.

Modern analytics often uses ELT.

Data lake

Raw/semi-structured storage, usually object storage.

In GCP:

Cloud Storage

Good for:

Raw files

Cheap storage

Flexible formats

Data science exploration

Data warehouse

Structured, query-optimized analytics system.

In GCP:

BigQuery

Good for:

SQL
Dashboards
Analytics
Business reporting

Lakehouse

A hybrid approach: data lake storage plus warehouse-like governance/querying.

In GCP:

Cloud Storage + BigLake + BigQuery + Dataplex

Federated query

Querying data where it lives, without moving/loading it first.

Examples:

BigQuery querying Cloud SQL
BigQuery querying Cloud Storage external table
BigQuery querying Google Sheets

Use when:

Data is small
Data must stay in source
You need quick exploration
You want to avoid duplication

Avoid when:

High performance is required
Data is huge and queried frequently
You need strong modeling/partitioning

CDC — Change Data Capture

Capturing inserts, updates, and deletes from a database.

Example:

Postgres order table changes → Datastream → BigQuery

Use for near-real-time replication.

Schema

The structure of your data.

Example:

```
customer_id STRING
order_id STRING
order_timestamp TIMESTAMP
amount NUMERIC
```

Partitioning

Splitting a large table by date/time or another key to reduce query cost and improve performance.

Example:

Partition orders table by order_date

Clustering

Organizing data within partitions by commonly filtered columns.

Example:

Cluster orders by customer_id or region

Data quality

Checks that data is correct, complete, and usable.

Examples:

No duplicate order IDs

No null customer IDs

Revenue cannot be negative

Row counts should not drop 90%

Data lineage

Tracking where data came from, where it went, and how it changed.

Example:

raw_orders → cleaned_orders → revenue_dashboard

Google describes lineage as tracking the data lifecycle from origin to destination and the transformations along the way. ([Google Cloud Documentation](#))

Orchestration

Coordinating pipeline steps.

Example:

1. Pull file
2. Load into BigQuery
3. Run transformation
4. Run quality checks
5. Refresh dashboard
6. Notify team

Tools:

Composer

Workflows

Cloud Scheduler

Dataform

Idempotency

A pipeline can run multiple times without creating bad duplicates or inconsistent output.

Example:

If today's pipeline reruns, it replaces today's partition instead of appending duplicates.

This is very important on the job.

Backfill

Running a pipeline for historical data.

Example:

Pipeline normally runs daily, but now you load the last 2 years.

SLA / SLO

Reliability expectations.

Example:

Dashboard data must be refreshed by 8 AM daily.

Pipeline must be 99.9% reliable.

Phase 4 — Your Timeline From Now Until Monday

Assuming you start Monday, June 29, the goal is not mastery of every command. The goal is **fluency**.

Wednesday — Tool map and BigQuery foundation

Focus:

BigQuery
Cloud Storage
External tables
BigLake
bq load
LOAD DATA
Data Transfer Service
Partitioning
Clustering

Outcome:

You should be able to explain where data lives, how it gets into BigQuery, and when to load vs query externally.

Mini-build:

Upload CSV to Cloud Storage
Load it into BigQuery
Create external table
Compare native vs external table
Create partitioned/clustered table
Run basic SQL

Thursday — Ingestion and processing

Focus:

Pub/Sub
Dataflow
Datastream
Data Fusion
Dataproc
ETL vs ELT
Batch vs streaming
CDC

Outcome:

You should know when to use Dataflow vs Dataproc vs Data Fusion.

Mini-build:

Simulate event data
Publish events to Pub/Sub

Process with Dataflow or simpler Cloud Function/Cloud Run first
Land data in BigQuery

Friday — Orchestration, governance, and reliability

Focus:

Composer
Workflows
Cloud Scheduler
Dataform
Dataplex
IAM
Data quality
Lineage
Monitoring
Logging

Outcome:

You should understand how production pipelines are scheduled, monitored, and governed.

Mini-build:

Schedule a BigQuery transformation
Add data quality checks
Create monitoring/alerting
Document lineage

Saturday — Google ADK and Agent Platform

Focus:

ADK
Agent Platform
Agent Studio
Agent Garden
Model Garden
Vertex AI
Tool-calling agents
Agents over BigQuery

Agent memory/sessions

Outcome:

You should be able to explain how an agent can help with data engineering workflows.

Mini-build:

Build an agent that answers questions from BigQuery

Give it tools:

- run SQL
- summarize results
- explain anomalies
- suggest pipeline fixes

Sunday — End-to-end real-world builds

Focus:

Full pipeline architecture

Debugging

Tradeoff decisions

Cost/security/performance

Interview/job explanations

Outcome:

You should be able to look at a real business problem and design the GCP architecture.

Mini-build:

Pick 1–2 of the real-world use cases below and build the basic version.

Phase 5 — Real-World Use Cases to Build

These are useful projects because they map to real data engineering work and they also let you experiment with agents.

Use Case 1 — Restaurant Analytics Pipeline for MenYou

This is the most personally useful one for you.

Business problem

Restaurants upload menus, reviews, item data, and order/click behavior. You want analytics on:

- Most viewed menu items
- Most recommended dishes
- User taste trends
- Restaurant performance
- Menu item popularity

GCP architecture

- Frontend/app events
 - Pub/Sub
 - Dataflow
 - BigQuery raw events
 - Dataform transformations
 - BigQuery curated tables
 - Looker Studio dashboard
 - ADK agent for business questions

Tools learned

- Pub/Sub
- Dataflow
- BigQuery
- Dataform
- Looker Studio
- ADK
- IAM
- Monitoring

Agent idea

Build a “Restaurant Insights Agent.”

It can answer:

Why did recommendations drop yesterday?

Which dishes are trending?
Which restaurant has low menu engagement?
What menu items should we promote?

Why this is good

This gives you a startup-relevant and job-relevant data project.

Use Case 2 — E-commerce Orders Data Warehouse

Business problem

A company wants to analyze orders, customers, products, revenue, and returns.

GCP architecture

CSV/API orders
→ Cloud Storage raw zone
→ BigQuery raw tables
→ Dataform SQL transformations
→ Star schema:
 fact_orders
 dim_customers
 dim_products
 dim_dates
→ Looker dashboard
→ Data quality checks

Tools learned

Cloud Storage
BigQuery
LOAD DATA
Dataform
Partitioning
Clustering
Looker Studio
Dataplex/lineage

Agent idea

Build a “Revenue Analyst Agent.”

It can answer:

Why did revenue change this week?

Which products drove growth?

Are returns increasing?

Which customer segment is most valuable?

Why this is good

This is the classic data warehouse project. Very useful for interviews and real work.

Use Case 3 — Real-Time Clickstream Pipeline

Business problem

A website wants to track user behavior in near real time.

GCP architecture

Website/app events

→ Pub/Sub

→ Dataflow streaming

→ BigQuery

→ Real-time dashboard

→ Monitoring alerts

Tools learned

Pub/Sub

Dataflow

BigQuery streaming

Windowing

Deduplication

Monitoring

Logging

Agent idea

Build a “Traffic Anomaly Agent.”

It can answer:

Did traffic spike?

Which page is causing errors?

Did conversion rate drop?
Is this a real anomaly or missing data?

Why this is good

This teaches streaming, which is a major data engineering concept.

Use Case 4 — Operational Database to Analytics using CDC

Business problem

A company has a PostgreSQL app database and wants analytics in BigQuery without manually exporting data.

GCP architecture

Cloud SQL PostgreSQL
→ Datastream CDC
→ Cloud Storage / BigQuery
→ BigQuery analytics tables
→ Dataform transformations
→ Dashboard

Tools learned

Cloud SQL
Datastream
CDC
BigQuery
Dataform
Data quality
IAM

Agent idea

Build a “Database Sync Agent.”

It can answer:

Is replication healthy?
What tables changed today?
Are source and BigQuery row counts matching?

Did the pipeline miss updates?

Why this is good

This is very realistic. Many companies need operational DB → warehouse replication.

Use Case 5 — Data Governance and PII Detection Project

Business problem

A company has customer data and needs to detect sensitive fields, restrict access, and track lineage.

GCP architecture

Customer data in Cloud Storage/BigQuery

- Sensitive Data Protection scan
- Policy tags / column security
- Dataplex catalog
- Lineage tracking
- Access-controlled BigQuery views

Tools learned

BigQuery security
IAM
Policy tags
Sensitive Data Protection
Dataplex
Lineage
Authorized views

Agent idea

Build a “Data Governance Agent.”

It can answer:

Which tables contain PII?
Who owns this table?
Can this dataset be shared?
What columns should be masked?

What downstream dashboards use this table?

Why this is good

This teaches the “production enterprise” side of data engineering.

The order I recommend for you

Given your timeline and job goal, do this order:

1. E-commerce Orders Data Warehouse
2. MenYou Restaurant Analytics Pipeline
3. Real-Time Clickstream Pipeline
4. CDC Database to BigQuery Pipeline
5. Governance/PII Project

The first two will give you the strongest foundation. The third teaches streaming. The fourth teaches database replication. The fifth teaches enterprise governance.

For now, our next step should be **Phase 1: the GCP tool map**, but we'll go one group at a time:

Group 1: Storage + warehouse
Cloud Storage, BigQuery, External Tables, BigLake

Group 2: Ingestion
bq load, LOAD DATA, BigQuery Data Transfer Service, Pub/Sub, Datastream

Group 3: Processing
Dataflow, Dataproc, Data Fusion, Dataform

Group 4: Orchestration
Composer, Workflows, Scheduler

Group 5: Governance/security
Dataplex, IAM, policy tags, DLP, lineage

Group 6: AI/agents
Vertex AI, ADK, Agent Platform

Start with **Group 1**: Cloud Storage vs BigQuery vs External Tables vs BigLake.

